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COLLEGE OF ENGINEERING, TECHNOLOGY, ARCHITECTURE
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BACHELOR OF SCIENCE IN COMPUTER ENGINEERING CPE
COGNATE/TRACK COURSE 1
CPEP 323 A

NEON DODGE: LEVEL OVERDRIVE

Hand-Tracking Arcade Dodge Game — Documentation

Submitted by:

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Introduction

Neon Dodge: Level Overdrive is a real-time hand-tracking arcade game built in Python using Pygame and MediaPipe. The player controls an on-screen avatar exclusively through physical hand gestures captured by a standard webcam — no keyboard, mouse, or gamepad required. The game is designed around fast, reflex-driven gameplay set across three escalating levels of difficulty.

The project demonstrates the integration of computer vision, game development, and real-time input processing into a cohesive, polished experience. It showcases how MediaPipe hand landmark detection can be leveraged as a novel and accessible game controller, translating physical movement into responsive in-game action.

This document provides a comprehensive technical and design overview of Neon Dodge: Level Overdrive, covering the game concept, mechanics, architecture, algorithms, user interface, and development process. It is intended for developers, evaluators, and anyone seeking a thorough understanding of how the project was conceived, designed, and implemented.

Objectives

The primary objectives of Neon Dodge: Level Overdrive are as follows:

- Design and implement a playable, fully functional arcade-style game using Python and the Pygame library.
- Integrate real-time hand tracking via MediaPipe to serve as the sole input mechanism, replacing traditional controllers.

- Create a multi-level progression system with increasing difficulty, featuring distinct visual themes and enemy behaviors per level.
- Implement a robust power-up system that modifies gameplay dynamically, including health recovery, damage immunity, and slow-motion effects.
- Deliver a polished audio-visual experience through per-level background music, dynamic sound effects, screen shake, chromatic aberration, cinematic transitions, and particle systems.
- Ensure smooth, responsive gameplay at 60 frames per second with proper letterboxing for fullscreen display across monitor resolutions.
- Provide clear in-game feedback through HUD elements, including health points, survival timer, and active power-up indicators.

Scope and Limitations

Scope

- A complete single-player game loop with three levels, a start menu, level intros, stage-cleared transitions, and a credits screen.
- Real-time hand tracking using a consumer webcam and MediaPipe, with gesture recognition for menu interactions.
- Five distinct enemy/hazard types: Exploders, Mini Circles, Lasers, Big Orbs (Black Hole), and Sun Orbs.
- Three power-up types: HP Boost, Shield, and Slow Motion.

- Dynamic audio management including per-level music, dedicated SFX channels, and graceful fade transitions.
- Screen effects including chromatic aberration, screen shake, zoom, vignette, scanlines, and particle systems.

Limitations

- The game requires a functional webcam for input. There is no keyboard/mouse fallback input mode.
- Performance of hand tracking may vary based on hardware capability, lighting conditions, and camera quality.
- The hand tracker is single-hand only (`max_num_hands=1`); multi-hand gestures are not supported.
- The game is designed for a 900x600 logical resolution, letterboxed to the native monitor resolution. Non-standard aspect ratios may display with wider letterbox bars.
- Game state is not saved or persisted between sessions. There is no pause function or settings menu.
- Asset files (images, audio) are required to exist in the local assets/ directory; missing files are logged but not gracefully recovered.

Game Overview/Concept

Neon Dodge: Level Overdrive is a survival arcade game set in a neon-lit cyberpunk world. The player's goal is simple: survive for 60 seconds on each of three increasingly chaotic levels while dodging a growing variety of enemies and hazards.

The game is controlled entirely by the player's index finger — pointing your finger at the camera moves the on-screen avatar toward that position. This creates a direct, physical mapping between real-world gesture and in-game movement that feels intuitive within seconds.

Aesthetic and Theme

Each level has a distinct color palette and musical identity:

Level	Name	Theme / Aesthetic
1	The Awakening	Cool blue; calm introduction, slow lasers
2	Pulse Overdrive	Purple; rapid lasers, sun orbs introduced
3	Chaos Protocol	Dark red; all hazards active, black hole appears

The visual style employs glowing neon circles, scrolling grid backgrounds, pulsing particles, and cinematic screen effects to create a high-energy, retro-futuristic atmosphere.

Win/Loss Condition

- Win: Survive for 60 seconds (3,600 frames at 60 FPS) on a level to clear it and advance.
- Loss: The player has 3 hit points. Losing all HP triggers a death animation and restarts the current level after a 3-second respawn delay.

Game Mechanics

Player Movement

The player avatar is a glowing circle that moves toward the position of the player's real-world index fingertip. The HandTracker module uses a linear interpolation (lerp) factor of 0.55 per frame, providing a balance between responsiveness and camera noise smoothing. A margin mapping of 15% on each edge of the camera frame is applied, meaning the player can reach screen edges without moving their hand off-camera.

Player Hit Points and Damage

The player starts each level with 3 HP. Upon taking damage, the player becomes briefly invulnerable (invulnerable frames) to prevent rapid multi-hit scenarios. Visual feedback includes a flash effect and chromatic aberration. When HP reaches 0, a death sequence plays (fade to black over 3 seconds with death music) before the level restarts.

Hazard Types

Hazard	Behavior	Notes
Exploder	Orbits the arena; detonates after a fuse timer, spawning Mini Circles.	Slow-motion extends fuse timer
Mini Circle	Fast-moving particles ejected from Exploder detonation.	Disappear on player contact
Laser	Fires from a random screen edge with a warning indicator, then sweeps across.	Speed increases per level
Big Orb (Black Hole)	Grows to massive size in Level 3; kills player if too close.	Managed manually in <code>dodge.py</code>
Sun Orb	Spawned in pairs; tracks toward the player.	Introduced in Level 2

Power-up System

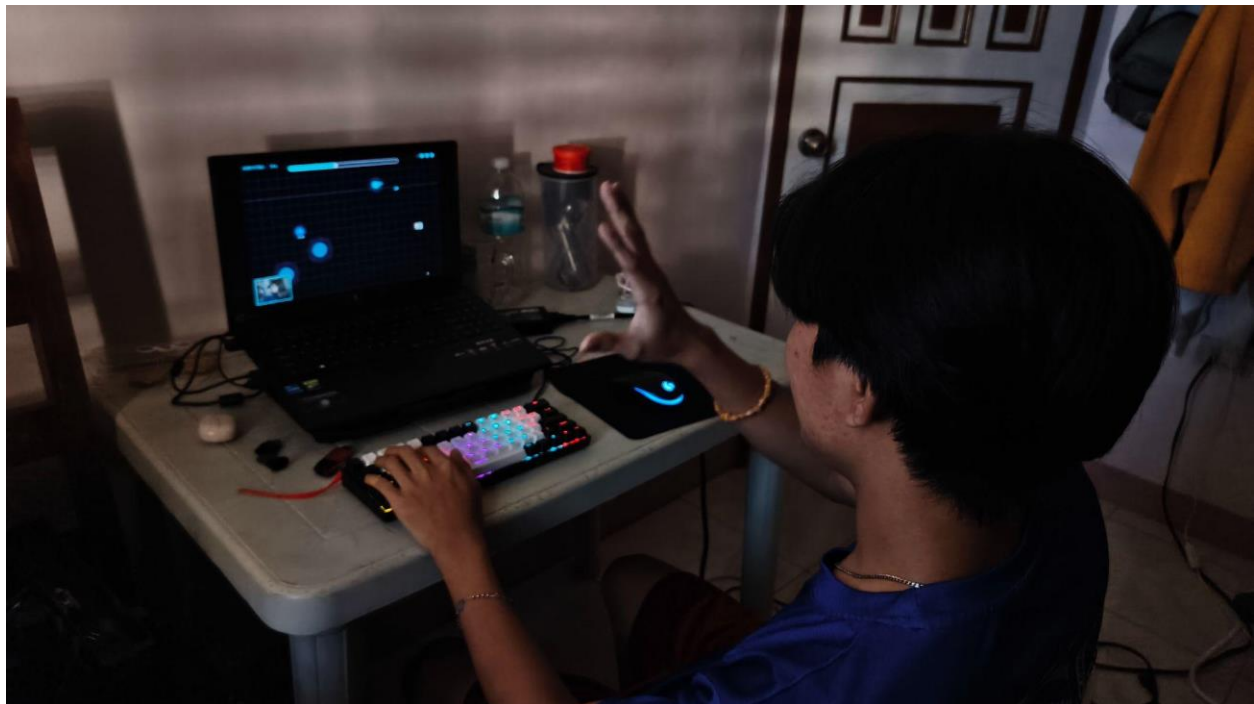
Power-ups spawn on the arena floor and must be touched to collect. Only one power-up is active at a time. They expire if not collected within 5 seconds.

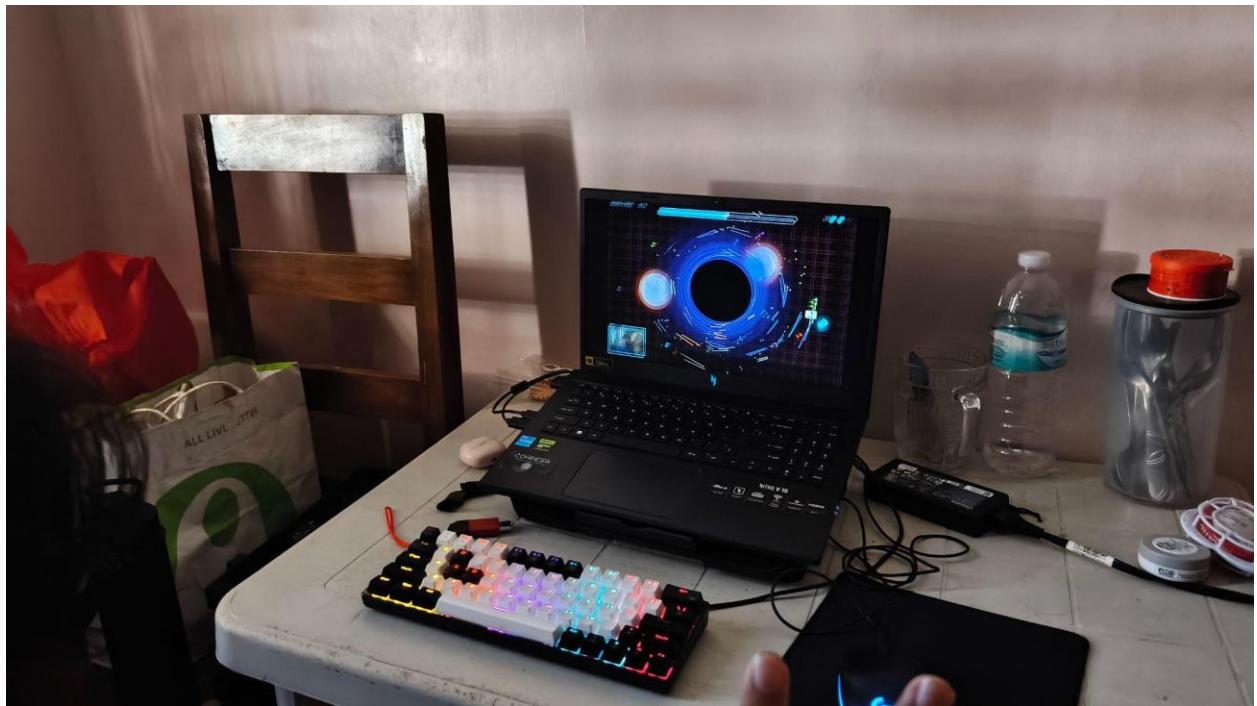
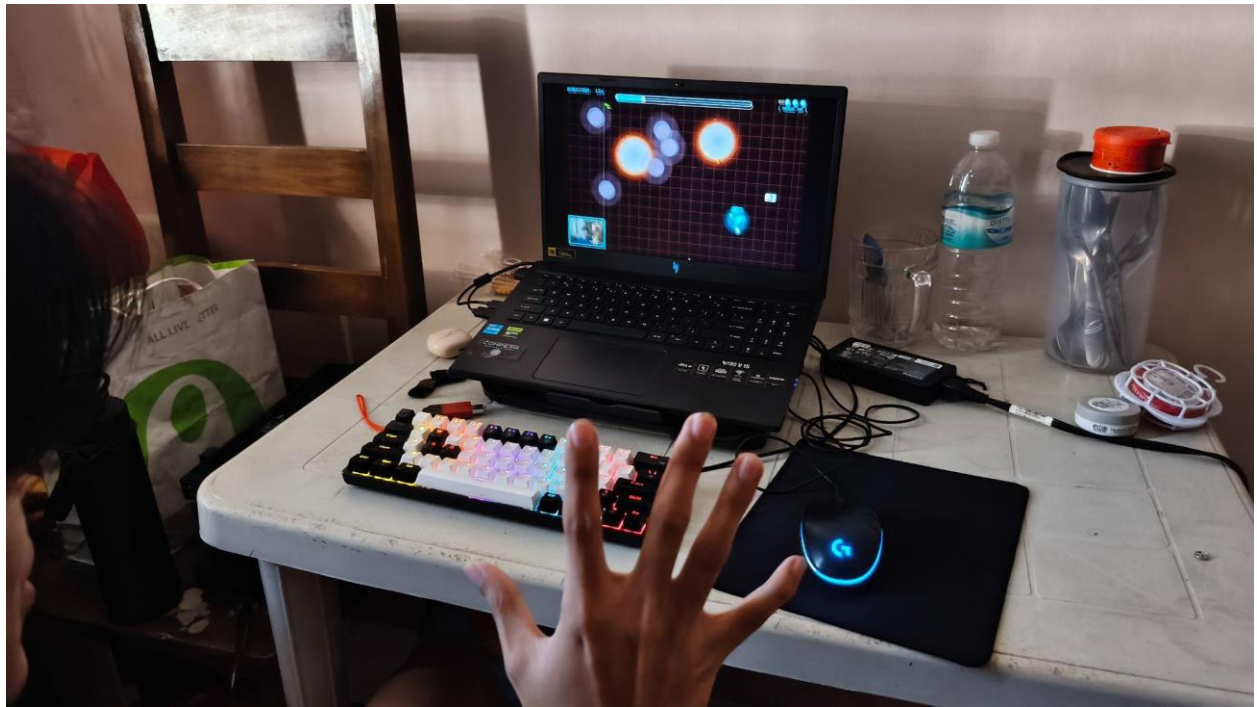
Power-Up	Effect	Duration
+HP	+2 temporary hit points (consumed on hit)	10 seconds or 2 hits
SHIELD	Absorbs all incoming damage completely	10 seconds

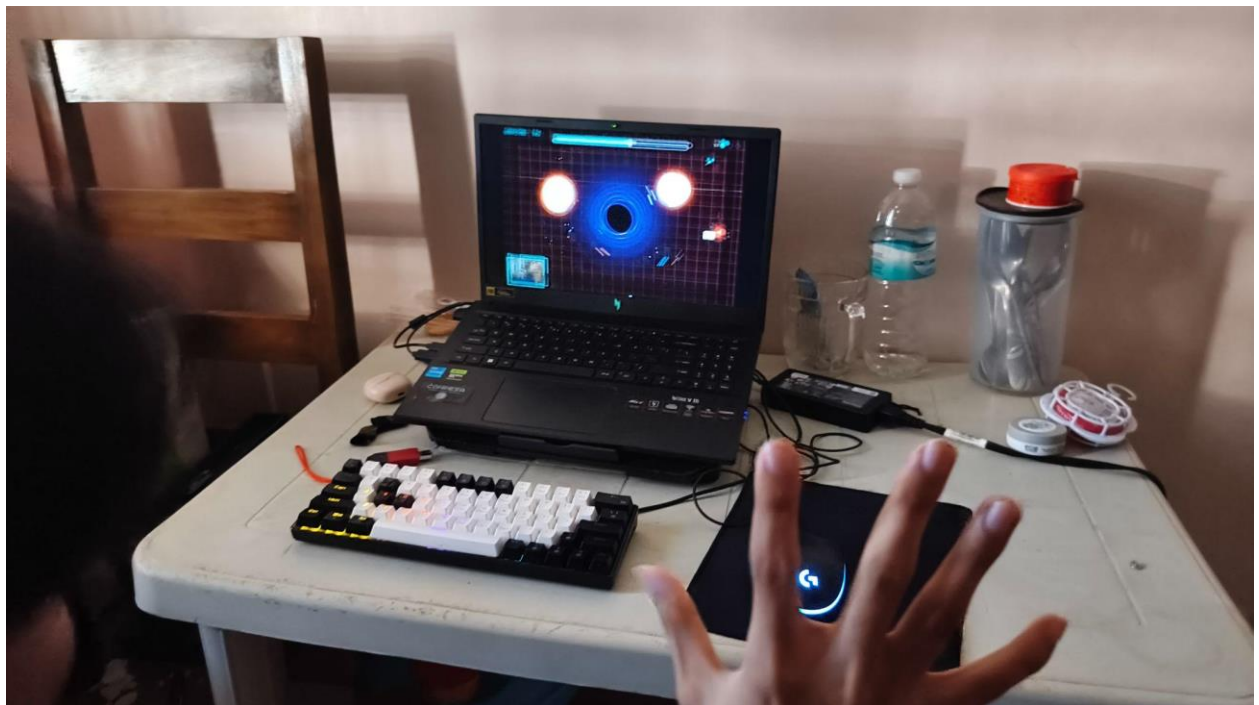
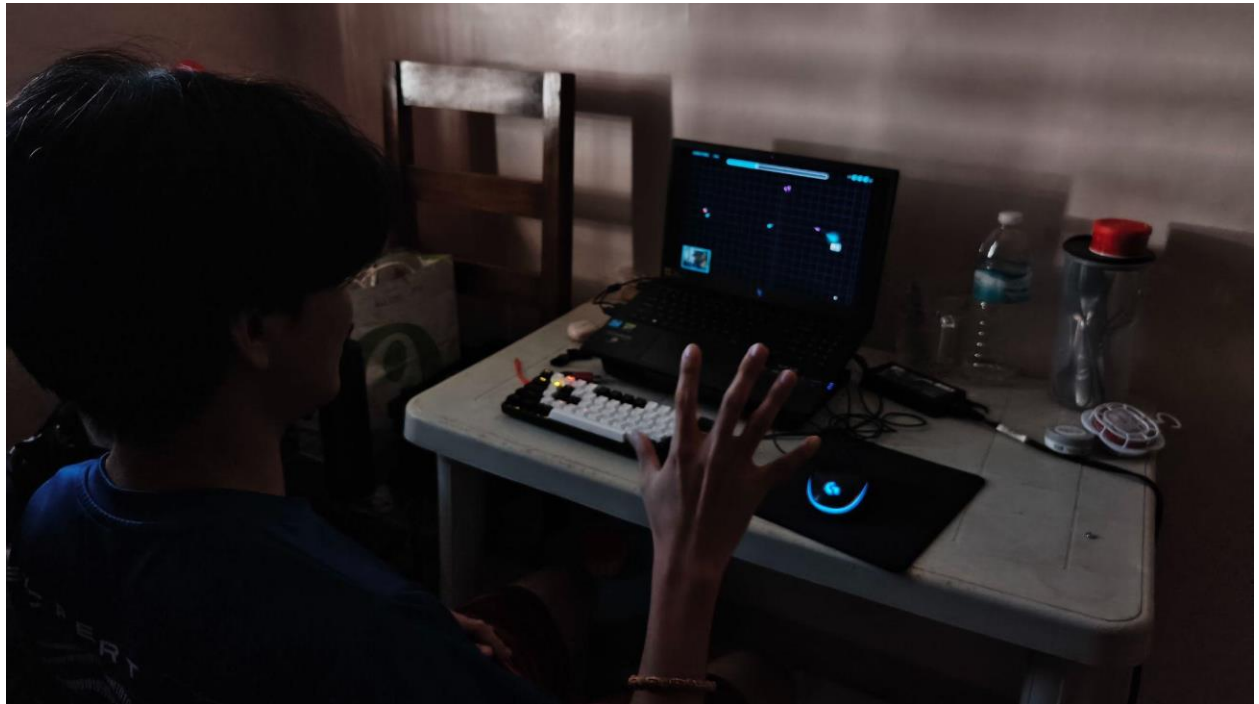
Power-Up	Effect	Duration
SLOW	Hazards move at 60% normal speed	10 seconds

Power-up spawn rules: first spawn within the first 10 seconds of each level; 15-second cooldown between pickups; only one on screen at a time.

Documentations







Conclusions

Neon Dodge: Level Overdrive successfully demonstrates the feasibility and appeal of using real-time computer vision as a primary game input mechanism. By combining MediaPipe hand tracking with Pygame's rendering capabilities, the project delivers a functional, polished arcade game that is controlled entirely through natural physical gesture.

The codebase is cleanly separated into four well-defined modules covering hand tracking, power-up logic, level configuration, and the main game loop. This separation of concerns ensures maintainability and makes the system easy to extend with new levels, hazard types, or power-ups.

The project meets all stated objectives: a playable game loop with three escalating difficulty levels, a novel and responsive input system, dynamic audio-visual feedback, and a complete game experience from start screen to credits. The identified limitations — primarily webcam dependency and single-hand tracking — represent practical engineering tradeoffs rather than fundamental design flaws, and clear paths for improvement are outlined in the Future Development section.

Neon Dodge: Level Overdrive stands as a complete demonstration of the intersection between game development, computer vision, and human-computer interaction design.

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Motto: Knowledge is power



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